doi:10.1038/nature11867

Supplementary Figures

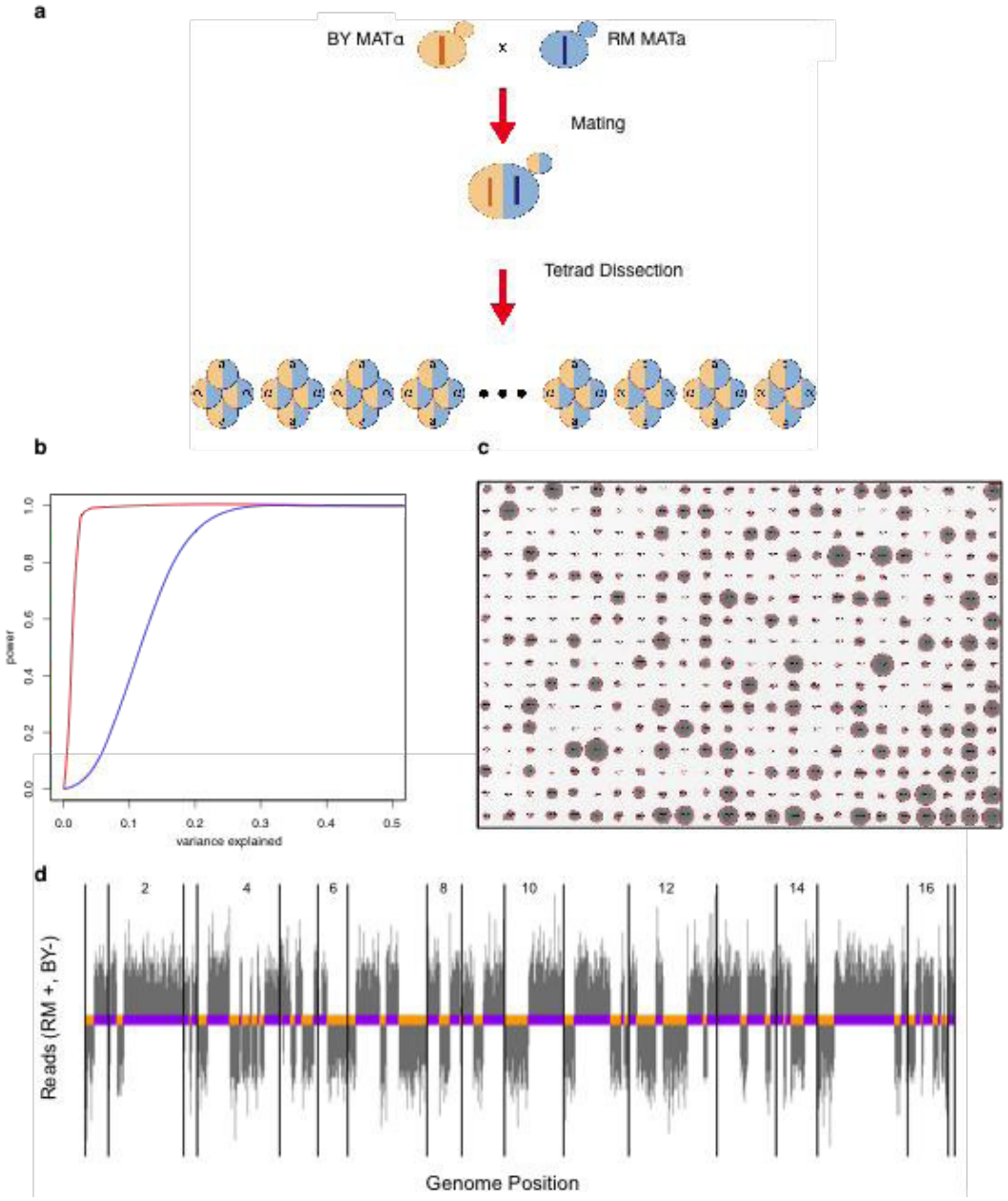


Figure S1. The design of the segregant panel is shown in (A). (B) Curves illustrating statistical power are shown for mapping populations of 100 (blue) and 1000 (red) segregants at a genome-wide significance threshold. (C) An image of endpoint colony growth is shown for 384 segregants, with the outlines of colonies, as detected by our image processing software, indicated in red. (D) Counts of sequencing reads at SNP sites are plotted (Y-axis) against genome position (X-axis) for a representative segregant; the orange (BY) and purple (RM) bars indicate parental haplotype calls, and the vertical black bars delineate chromosomes.

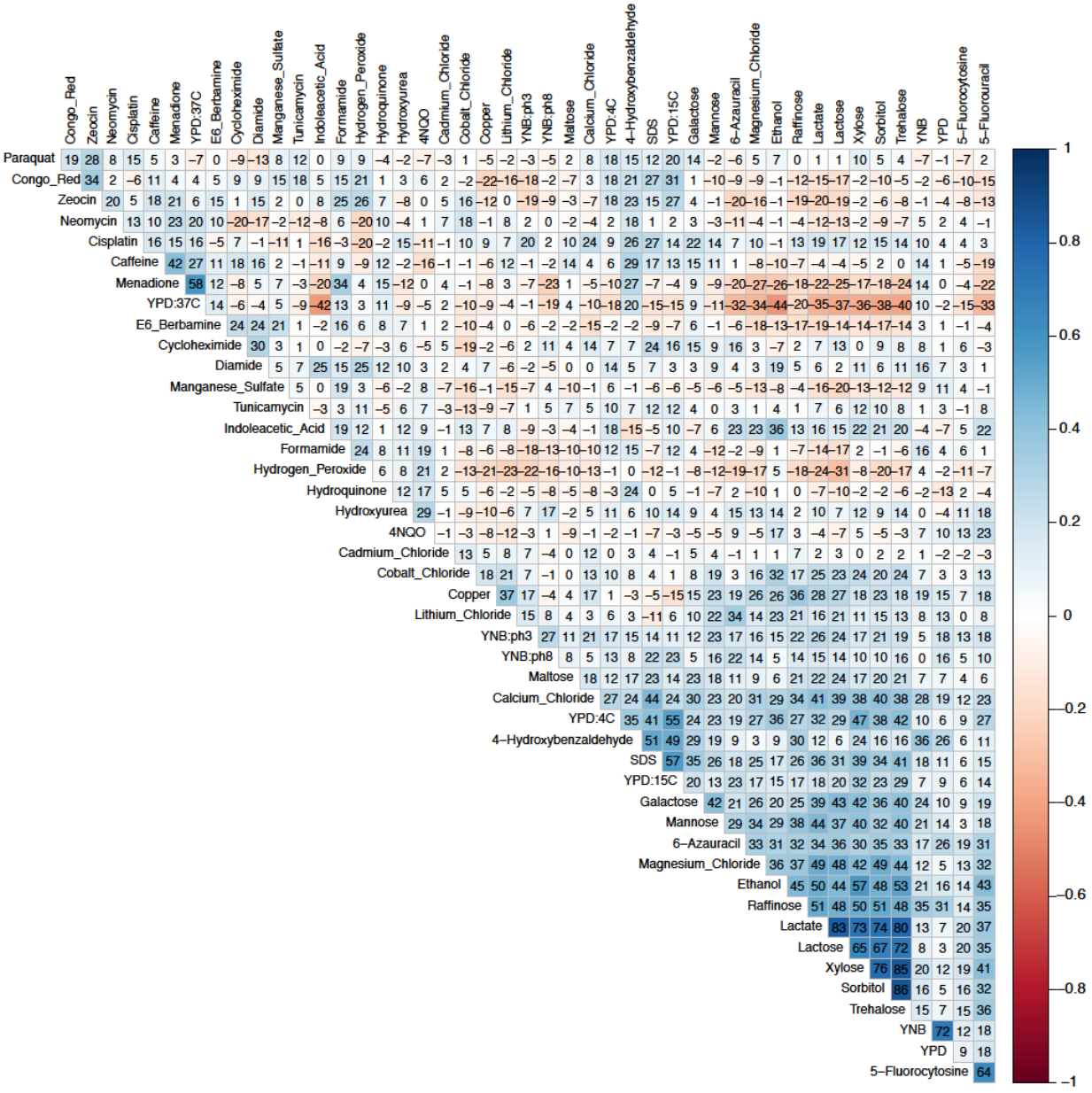


Figure S2. Spearman correlation coefficients for all pairs of traits are shown. Numbers in table cells indicate (100 * correlation coefficient).

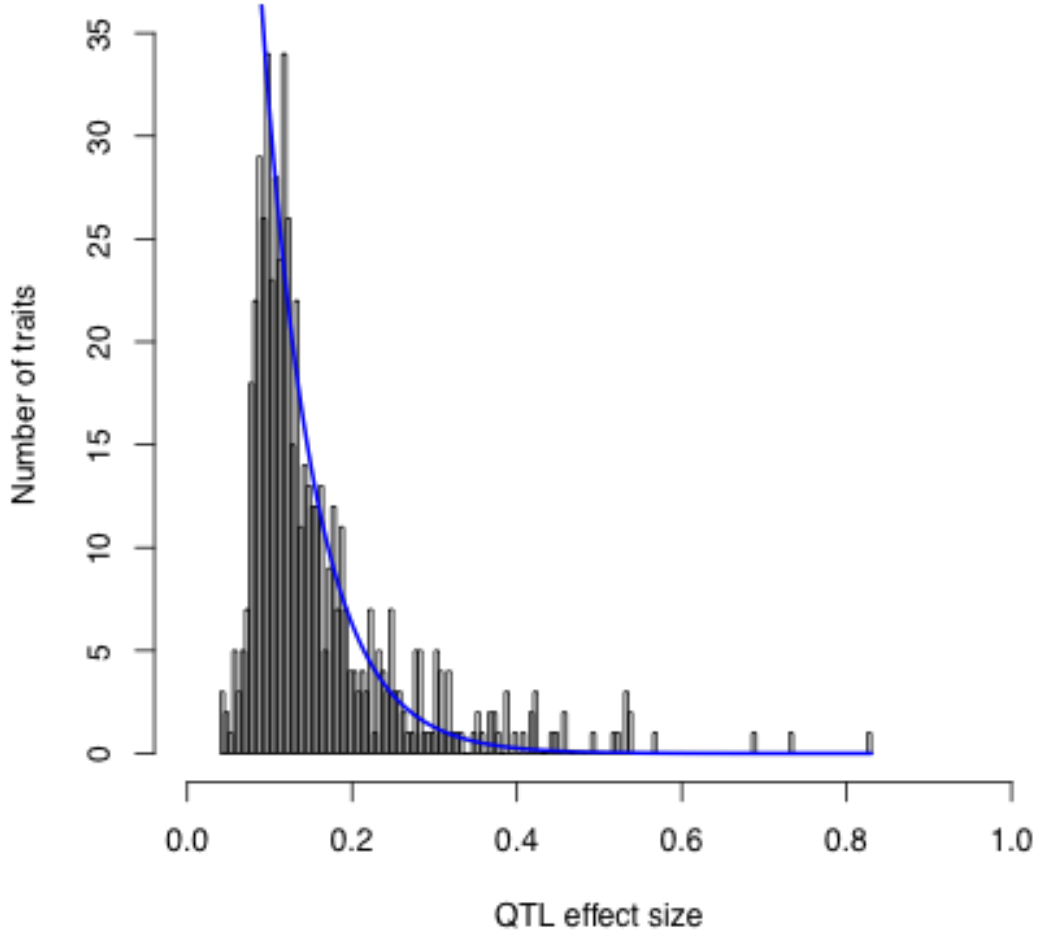


Figure S3. A histogram of QTL effect sizes across all traits is plotted, showing that most detected QTL have small effects. Effect size here is the absolute value of the standardized difference in allelic means for each QTL. The blue line indicates a fit of a truncated exponential distribution of effect sizes.

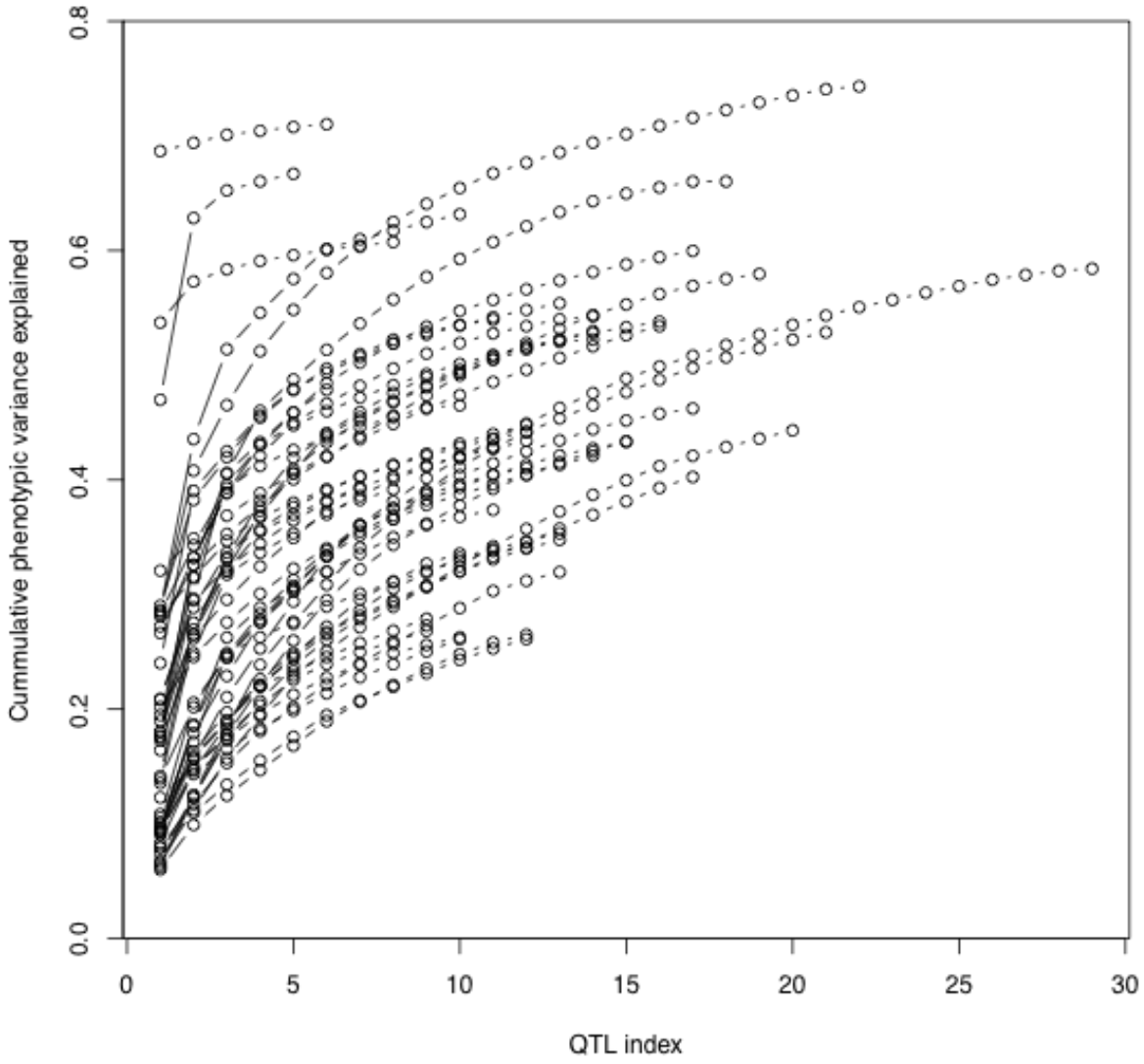


Figure S4. For each trait, the effect sizes of detected QTL are sorted from largest to smallest, and the cumulative phenotypic variance explained is plotted (Y-axis) against the number of detected QTL (X-axis). Under Orr's model, the curves should extrapolate to the origin; deviations from this expectation imply the presence of large-effect QTL for the corresponding traits.

Additional data and code can be found at:

http:/genomics-pubs.princeton.edu/YeastCross_BYxRM/